1 Z, L, M Matrices

- A Z**-matrix** has off-diagonal entries les than or equal to zero:

$$Z = (z_{ij}); \quad z_{ij} \leq 0; \quad i \neq j.$$

Here is an example:

$$\begin{pmatrix} 1 & -2 & 0 \\ 0 & 5 & -1 \\ -1 & -3 & -4 \end{pmatrix}.$$

- An L**-matrix** has off-diagonal entries less than or equal to zero, and diagonal entries are strictly positive:

$$L = (\ell_{ij}); \quad \ell_{ii} > 0, \quad \ell_{ij} \leq 0 \quad (i \neq j)$$

Here is an example:

$$\begin{pmatrix} 1 & -2 & 0 \\ 0 & 5 & -1 \\ -1 & -3 & 4 \end{pmatrix}.$$

- Finally, an M**-matrix** is defined as follows. A Z-matrix is a square matrix of form $A = \lambda I - P$, where P is nonnegative. Such a Z-matrix A is an M**-matrix** if $\lambda \geq \rho(P)$, where ρ denotes the spectral radius, $\rho(P) = \max\{|\mu| : \mu \in \sigma(P)\}$ and $\sigma(P)$ denotes the spectrum (set of eigenvalues) of P.

Here is an example. Let

$$P = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \geq 0$$

The spectral radius is $\rho(P) = \sqrt{2}$. Choosing $\lambda = 3 \geq \rho(P)$ yields the M-matrix:

$$A = \lambda I - P = \begin{pmatrix} 3 & -1 \\ -2 & 3 \end{pmatrix}$$

with least real eigenvalue:

$$l(A) = \lambda - \rho(P) = 3 - \sqrt{2} > 0$$